

Beyond Problem Solving: A Three-Space Probe of LLM Critical Thinking

Anonymous ACL submission

Abstract

Large language models (LLMs) are widely used for complex reasoning tasks, with various datasets designed to evaluate their problem-solving capabilities across different domains. However, limited attention has been given to LLM performance on unsolvable questions caused by inconsistencies. This paper proposes an evaluation framework grounded in Hierarchical Three-Space Theory to assess LLMs' critical thinking capabilities, specifically their ability to identify problem inconsistencies and challenge flawed problem setups. We design two types of inconsistency: insufficient information that renders answers indeterminate, and redundant information that introduces contradictions. We apply these modifications to multiple-choice problems spanning different fields and generative math problems. We establish detailed behavioral categories for LLM responses and analyze whether LLMs recognize inconsistencies and how they respond to them. Our experiments with state-of-the-art LLMs reveal significant limitations in critical thinking, demonstrating that critical thinking represents a distinct capability that cannot be improved solely through enhanced reasoning or problem-solving skills. Through comprehensive experiments, we analyze how prompting strategies and problem difficulty influence critical thinking performance.

1 Introduction

As large language models (LLMs) become increasingly integrated into decision-making processes, ensuring they possess robust critical thinking skills is of paramount importance. While significant attention has been given to LLMs' ability to generate responses and solve problems, the research community has also recognized the importance of understanding the limitations and potential risks associated with these models (Weidinger et al., 2022; Kaddour et al., 2023). Real-world user queries frequently contain inconsistencies, such as missing

essential information required to derive answers or including contradictory elements within the context. A crucial question arises: *Can LLMs critically assess the very foundation of a problem—its initial framing—and identify inherent inconsistencies?* Failure to do so could lead to flawed reasoning, inaccurate conclusions, and ultimately, unreliable performance, especially in complex, real-world scenarios.

Recent research has explored various facets of critical thinking in AI, including handling incomplete or ambiguous requests (Asai and Choi, 2021; Kamath et al., 2020; Kuhn et al., 2022), discerning truth from falsehood (Xu et al., 2023; Chen and Shu, 2023), and reconciling contradictory information (Xie et al., 2023; Zhou et al., 2023). However, the ability to recognize inconsistencies in problem framing remains under-explored. Current evaluation methods, while providing valuable insights into model performance on well-defined tasks, often fail to capture the challenges posed by such inconsistencies. This limitation highlights a significant gap in our understanding of LLMs' capabilities.

This study contributes to the ongoing discussion of LLM capabilities by introducing a novel framework for assessing this specific aspect of critical thinking in problem-solving. We contribute a general methodology for creating benchmarks to assess this crucial skill. Our work is grounded in the Three-Space Theory of Problem Solving (Burns and Vollemeyer, 2000), which describes problem-solving as a process of interacting searches within three distinct but interconnected spaces: the Problem Framing Space, the Strategy Space, and the Implementation Space. In question-answering(QA) tasks, the Problem Framing Space determines the model's understanding of the question and guides strategy selection. For example, when an LLM recognizes a multiple-choice question, it may adopt a strategy of analyzing each option to select the

most suitable answer. The initial assumption—that a correct answer exists among the options—can be updated if the LLM determines that no option is correct. Based on this theory framework, we define the critical thinking capability:

DEFINITION OF CRITICAL THINKING FOR LLMs. Building upon the Three-Space Theory, critical thinking for LLMs is the ability to analyze the *Problem Framing Space* and recognize flaws in its definition, potentially by leveraging the *Strategy* and *Implementation Spaces*.

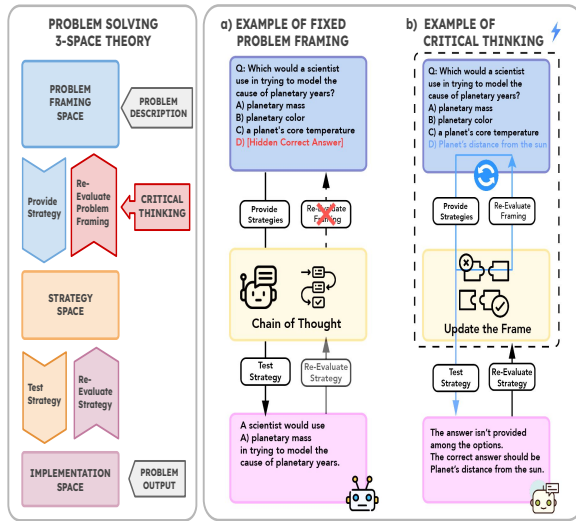


Figure 1: The Hierarchical Three-Space Theory of Problem-Solving illustrates the interplay between Problem Framing, Strategy, and Implementation Spaces. Critical thinking involves recognizing flaws in one’s understanding of a problem and leveraging feedback from the problem-solving process. The multiple-choice example illustrates this: an LLM, despite possessing relevant knowledge, might be constrained by a flawed problem setup, leading it to select an incorrect option and fabricate an explanation. However, an LLM with critical thinking capabilities would identify the issue and challenge the implausible options.

In this paper, we present a series of experiments designed to evaluate critical thinking in LLMs, focusing on their ability to recognize inconsistencies in problem framing. We introduce two types of modifications to create inconsistent, unsolvable questions from well-defined problems: insufficient information and redundant information. We apply these modifications to both multiple-choice and generative math problems, then observe how LLMs attempt to solve these problems and respond to the inherent inconsistencies. Our contributions are as follows:

- We define LLM critical thinking capability through the lens of Three-space problem-solving theory, establishing a crucial metric for evaluating LLM performance on inconsistent questions. We design an evaluation framework to analyze LLM behavior based on the Three-space theory.
- We develop a method to introduce inconsistencies, including insufficient information and redundant information, into well-defined questions. We get a comprehensive evaluation of prevalent LLMs’ critical thinking capabilities on the modified questions.
- We propose hypotheses explaining the observed differences in critical thinking performance across various scenarios and identify potential factors influencing current LLM critical thinking capabilities.

2 Related Work

Problem-Solving in Cognitive Science The Hierarchical Three-Space Theory of problem-solving is grounded in classic cognitive science theories (Newell, 1972; Stein et al., 1984) and addresses challenges of ill-structured problems (Rittel and Webber, 1973; Simon, 1973). Its dynamic Problem Framing Space aligns with metacognitive processes (Flavell, 1979) and complex problem-solving research (Dörner, 1986; Funke, 2010; Greiff et al., 2014), representing interactions between problem framing, strategy development, and implementation. The theory integrates critical thinking skills (Elder and Paul, 2007; Facione, 1990; Dwyer et al., 2014) and resonates with current complex problem-solving (CPS) frameworks (Quesada* et al., 2005; Grable, 2006). Our framework extends these foundations, offering complementary perspectives for evaluating LLMs. It provides a structured approach to assess critical thinking in artificial agents, introducing quantifiable metrics evaluating the LLM response behavior pattern. Our framework examination of interactions between Problem Framing, Strategy, and Implementation spaces offers a novel lens for understanding complex problem-solving processes. The LLM problem-solving strategy is shaped by prompting techniques. Chain-of-Thought (CoT) (Wei et al., 2022) breaks down problems into intermediate reasoning steps. Tree-of-Thought (Yao et al., 2024) extends CoT by exploring multiple branches of reasoning through

a tree structure. Graph-of-Thought (Besta et al., 2024) extends CoT by structuring the reasoning process as a graph. Algorithm-of-Thought (Sel et al., 2023) provides well-defined rules to guide the LLMs to reason logically and effectively. By focusing on LLMs’ robustness to misinformation and the influence of model architecture on problem-solving capabilities, our work addresses contemporary challenges in AI.

Critical Thinking in AI Recent literature explores critical thinking in AI through various lenses, including LLM noncompliance (Asai and Choi, 2021; Kamath et al., 2020; Brahman et al., 2024), misinformation susceptibility (Xu et al., 2023; Chen and Shu, 2023), knowledge conflicts (Xie et al., 2023; Zhou et al., 2023), input perturbations (Jia and Liang, 2017; Zhao et al., 2021), and sycophancy (Perez et al., 2023; Wei et al., 2023). These studies examine various facets of critical thinking in LLMs, including their ability to recognize limitations, handle misinformation, resolve contradictions, and resist biases. An emerging trend focuses on evaluating LLMs’ ability to assess and correct reasoning processes, as exemplified by benchmarks like MR-BEN (Zeng et al., 2024), PRM800K (Lightman et al., 2023), and MR-MATH (Xia et al., 2024), and others that evaluate higher-order cognitive skills by examining the reasoning process. Recent work has further explored LLMs’ capacity for self-correction (Tyen et al., 2023; Huang et al., 2023) and provided metrics for scoring step-by-step reasoning (Golovneva et al., 2023).

Our work distinguishes itself by focusing on LLMs’ capacity to critique problem formulations across domains and actively identify flaws in problem setups, a fundamental aspect of critical thinking often overlooked. Using the Three-Space Theory, we provide a unified framework to evaluate this ability in multiple-choice and mathematical problems, offering a comprehensive, cross-domain analysis of this critical thinking skill. Compared with existing benchmarks evaluating LLM on ambiguous or unanswerable questions (Brahman et al., 2024; Tian et al., 2023; Min et al., 2020), we create our dataset by modifying the options or context of well-defined questions and provide a more fine-grained analysis of LLM responses, investigating the factors that influence their critical thinking capabilities.

3 Critical Thinking in LLMs

3.1 Hierarchical Three-Space Theory for LLMs Critical Thinking

We adapt the Hierarchical Three-Space Theory (visualized in Fig. 1) to the context of language model processing, reframing the three spaces as:¹

Problem Framing Space (Model Space): Represents the LLM’s initial understanding and assumptions about the given task/question, derived from the prompt and the model’s pre-trained knowledge.

Strategy Space (Hypothesis Space): Encompasses potential reasoning paths or approaches to address the task, manifesting in the model’s generation of intermediate thoughts or steps, such as those observed in chain-of-thought reasoning. The exploration of this space is influenced by the model’s training and the specific prompting technique used.

Implementation Space (Experiment Space): Represents the actual output generation process, including token-by-token text generation where the model applies its selected strategy to produce a response. This space is directly observable through the model’s output.

Grounded in the Three-Space Theory, we analyze LLM critical thinking as an ability to facilitate feedback from the Strategy and Implementation Spaces to enable understanding update in the Problem Framing Space. For QA tasks, the Problem Framing Space represents the LLM’s understanding of the question and determines available actions. Well-defined multiple-choice questions limit actions to selecting a final choice. Actions like "none of the above" or "the answer cannot be determined" after correct reasoning indicate the LLM has identified question inconsistencies and updated its Problem Framing Space understanding.

3.2 Benchmark Creation Overview, Reproduction, and Experimental Setup

Our work establishes a framework to evaluate large language models’ critical thinking abilities by modifying existing, correctly annotated datasets. We conduct evaluations through multiple-choice problems and generative tasks, introducing two types of inconsistencies that render questions definitively unsolvable: insufficient information and redundant information. Insufficient information

¹We have renamed the spaces from the original Three-Space Theory (shown in parentheses) to better reflect their application to LLMs and avoid terminological confusion (e.g., ‘hypothesis space’ or ‘model’ in ML).

means removing crucial elements from the problem context, while redundant information introduces contradictions between the problem context and ground-truth knowledge or inside the problem context. These modifications enable us to assess a model’s ability to recognize inconsistencies and challenge inconsistent problem setups. Our evaluation datasets span diverse problem domains, including mathematics, physics, medical, science, and logical inference, each designed to assess specific problem-solving skills. We prioritize reasoning tasks as they align with our definition of critical thinking while providing observable intermediate steps.

Datasets. TAL(math)(matheval.ai, 2023), ARC Challenge (science)(Clark et al., 2018), MMLU college medicine, MMLU college physics, MMLU high physics(Hendrycks et al., 2020), BBH date understanding (date inference), BBH logical 3-objects, BBH logical 5-objects(logical order inference)(Suzgun et al., 2022), GSM8K(Cobbe et al., 2021) and SVAMP(generative math)(Patel et al., 2021). For each dataset, we sample up to 300 test queries for evaluation.

Insufficient Information.

- **(Hidden Correct Option):** For 8 multiple-choice datasets, we remove the correct answer choice from the option pool. Here, we study whether the model is able to update its Problem Framing assumptions that the correct answer choice might actually not be provided within the problem statement.
- **(Removing Necessary Condition):** For 2 generative math datasets, we remove one necessary condition from the problem statement so that the answer cannot be inferred from the provided context, thus, requiring the LLM to update its Problem Framing Space assumptions that the model cannot arrive at the final answer due to missing information.

Redundant Information. For multiple-choice problems, we add 3 types of misleading hints after the option, which provide counterfactual information about the problem.

- **(Gaslight Correct):** By the end of each problem statement, we add a hint claiming that the correct answer (e.g., *B*) is incorrect (e.g., *Hint: B is incorrect*).
- **(Gaslight Wrong):** By the end of each problem statement, we add a hint claiming that the

wrong answer (e.g., *A*) is correct (e.g., *Hint: A is correct*).

- **(Gaslight Both):** By the end of each problem statement, we add a hint claiming that the wrong answer is correct and the correct answer is incorrect (e.g., *Hint: A is correct and B is incorrect*).
- **(Redundant Condition):** We add additional information that contradicts the original problem context. We provide a numerical value for one intermediate variable. which is not consistent with the initial condition(e.g., *Context: Tom spends 2 on T-shirt and 4 on shoes. Hint: Tom totally spends 3 dollars.*)

We query GPT-4o to generate modified math questions and rephrase them coherently. We then verify that the numerical condition changes from original to modified questions satisfy our design requirements. Template illustrated in Appendix A.

Models. After constructing the evaluation datasets, we evaluate LLMs across various parameter sizes, capabilities, and reasoning strengths. The Qwen3(Yang et al., 2025) series models are specifically trained with controllable reasoning capabilities, allowing us to toggle reasoning/thinking mode on or off. We include the following models: GPT-4o, Llama3.3-70B (llama33_70)(Dubey et al., 2024), DeepSeekR1-distilled-Llama3.3-70B (dsr1_70), Qwen3-32B (qwen3_32), and Qwen3-8B (qwen3_8) with and without reasoning (thinking) mode. All models are configured with temperature 0 to ensure deterministic responses. For more details C.

Evaluation Metrics.

- **Problem-Solving Rate:** Measures whether the LLMs incorporate the correct knowledge about the question. We evaluate LLM accuracy on original well-defined questions. If an LLM can solve the clear question, we assume it possesses the required knowledge and correct solving approach.
- **Applied Critical Thinking Rate:** Measures the LLM’s ability to identify inconsistency in real-world application scenarios. When provided with user input that may be either clear or inconsistent, we expect the LLM to accurately classify the question type and generate an appropriate response. We define true-positive cases as responses exhibiting critical

behavior when the question is indeed inconsistent, and false-positive cases as instances where the LLM challenges well-defined problems. We calculate the precision p and recall r based on these values and use the F1 score: $\frac{2 * p * r}{p + r}$.

- **Inherent Critical-Thinking Rate:** Measures the LLM’s ability to identify problem inconsistency during correct reasoning processes. Unlike applied critical thinking, inherent critical thinking focuses on LLM behavior patterns after recognizing inconsistency, thus eliminating the performance discrepancy caused by problem-solving capability. According to Three-Space Problem Solving Theory, critical thinking influences the model’s choice when facing inconsistency: whether to trust its own knowledge and reasoning results or the problem setup. We first identify well-defined questions that the LLM solves correctly, then calculate the critical behavior ratio on corresponding modified versions. This approach ensures the LLM possesses the correct knowledge to solve the problem and can derive reasoning results that contradict the flawed question setup.

We develop the behavior categories for different responses and employ GPT-4o to label each response based on the defined category. According to the Three-Space framework, behavior types describe whether the reasoning process recognizes inconsistency and how the response reacts to it, whether the LLM challenges the problem setup based on reasoning results or revises previous solutions to fit the question setup. The designed category template reaches around 90% accuracy on our manually curated validation datasets, respectively. The category design for the different question types and inconsistency types, and validation performance are illustrated in the Appendix B.

3.3 Influential Factors for LLM Critical Thinking

The user query, expressed through specific prompt design, comprises two essential components: problem description and instruction. The problem description refers to the specific task at hand, while the instruction encompasses supplementary information about the problem and solution approach, like in-context examples. Both components influence understanding in the Problem Framing Space

and Strategy Space. The problem description establishes the model’s assumptions in the Problem Framing Space and helps the LLM infer appropriate strategies. Meanwhile, the instruction shapes the LLM’s strategic approach to the given problem and provides additional information within the Problem Framing Space. Based on these interactions, we propose two hypotheses to analyze LLM critical thinking performance and explain the relationship between the prompt and critical thinking.

Strategy Space Interaction (SSI) Hypothesis: The strategy an LLM employs to solve problems directly influences its ability to update its Problem Framing Space. The LLM’s problem-solving approach determines the reasoning path, which demonstrates how the LLM derives its final answer and signals whether each inference step is logical and convincing. Consider an extreme case: if an LLM generates only final choices without explanation, detecting flawed reasoning becomes impossible. The LLM’s strategy depends on both problem description and instructions. For any problem, the LLM starts with an initial strategy from pre-trained knowledge, then uses in-context learning to mimic instructional behavior patterns.

Inconsistency Difficulty (ID) Hypothesis: When an LLM’s solving process encounters inconsistency, it faces two basic choices: trust its own generation or assume the problem setup is incomplete. For easier tasks with which the LLM is more familiar or obviously counterfactual hints, it will show greater confidence in its generation and challenge the problem setup. The LLM’s critical thinking capability is influenced by question complexity and inconsistency obviousness.

4 Experiments and Results

Now, we present our analysis on each experiment delineated in Section 3 and study the relation to critical thinking ability.

4.1 Critical Thinking and Problem-solving

We use the scatter plot to present the overall performance of different models and datasets. Figures 2, 3 illustrate the relationship between problem-solving and critical thinking capabilities using accuracy and critical behavior ratios. For insufficient modification, the correlation coefficient between each model’s accuracy and critical behavior ratio across datasets, yielding $r = 0.258$ with p-value 0.0206 for applied critical thinking and $r = 0.225$

with p-value 0.0448 for inherent critical thinking. For redundant modification, the correlation coefficients are 0.175 and 0.280. This indicates only a weak positive correlation between problem-solving and critical thinking capabilities. The ability to challenge inconsistencies is not solely dependent on problem-solving proficiency. The accuracy of most of the models are above 0.75, but the critical score can vary between 0.3 to 0.8. Additionally, the scatter plot distribution in Figure 2 shows that generative math problems (GSM8k and SVAMP) and logical reasoning tasks (BBH logical3, and BBH logical5) typically exhibit higher critical behavior rates. This pattern likely occurs because inconsistencies are more apparent in these question types or have a more direct impact on the solving process to derive answers. In generative math problems, LLMs familiar with correct solving procedures can readily identify when necessary conditions are missing. For multiple-choice logical reasoning tasks (BBH logical3, and BBH logical5), LLMs can derive correct arrangement orders of the objects from the problem context, making it easier to detect invalid options. Compared to other multiple-choice problems, tasks requiring multi-step reasoning make inconsistencies more apparent than those relying primarily on memorized knowledge.

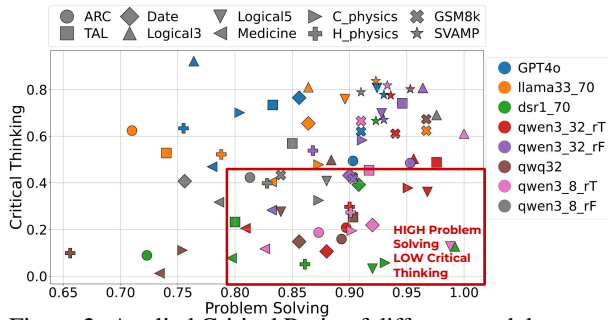


Figure 2: Applied Critical Ratio of different models across different datasets under insufficient information

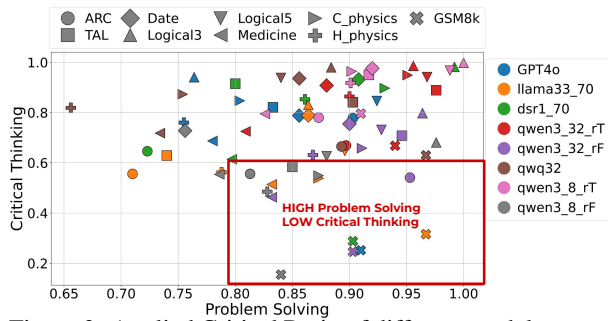


Figure 3: Applied Critical Ratio of different models across different datasets under redundant information.

4.2 Critical Thinking and Reasoning

We compare the critical behavior ratios of reasoning-enabled and base models of equivalent sizes. According to Figure 4, while facing insufficient information modification, non-reasoning models typically exhibit higher critical behavior rates, except for 32B and 8B models on generative math questions. For redundant information modification, all reasoning models outperform non-reasoning models. LLMs with reasoning capabilities are trained to solve problems logically through multi-step inference, typically producing lengthy Chain-of-Thought (CoT) responses. This training enhances performance when dealing with explicit inconsistencies that clearly obstruct answer derivation. Removing the necessary condition explicitly makes the answer undetermined. The redundant information create explicit contradiction, like misleading hints that contradict their derived answers, the reasoning capability helps them better conclude that the problem is ill-defined. However, modifications involving removed ground-truth options affect the answer derivation process more subtly. In these cases, reasoning capability may encourage LLMs to provide support or explanations for incorrect options that appear plausible under certain assumptions, even when such supporting information is not provided in the question context.

4.3 Problem Instruction Influence Critical Thinking

The Strategy Space defines how the LLM solves the problem, with different strategies leading to distinct reasoning paths. LLM predicts the next token based on its previous generation. Therefore, the final conclusion is sensitive to the chosen reasoning path. For any question, the LLM has initial understandings, and the strategy space determines available actions. LLM strategy is influenced by both question content and instruction. We tested four different instruction strategies: *basic*, *format*, *3-ICL*, and *3-ICL-cha*, with detailed prompting templates provided in Appendix A. The *basic* instruction imitates the most common user query by asking for the final answer and explanation. The *format* instruction provides a structured solving process including <question understanding>, <choice analysis> for multiple-choice problems or <intermediate value calculation> for math problems, <reasoning evaluation>, and <conclusion>. We designed this formatted problem-solving process to encourage the LLM

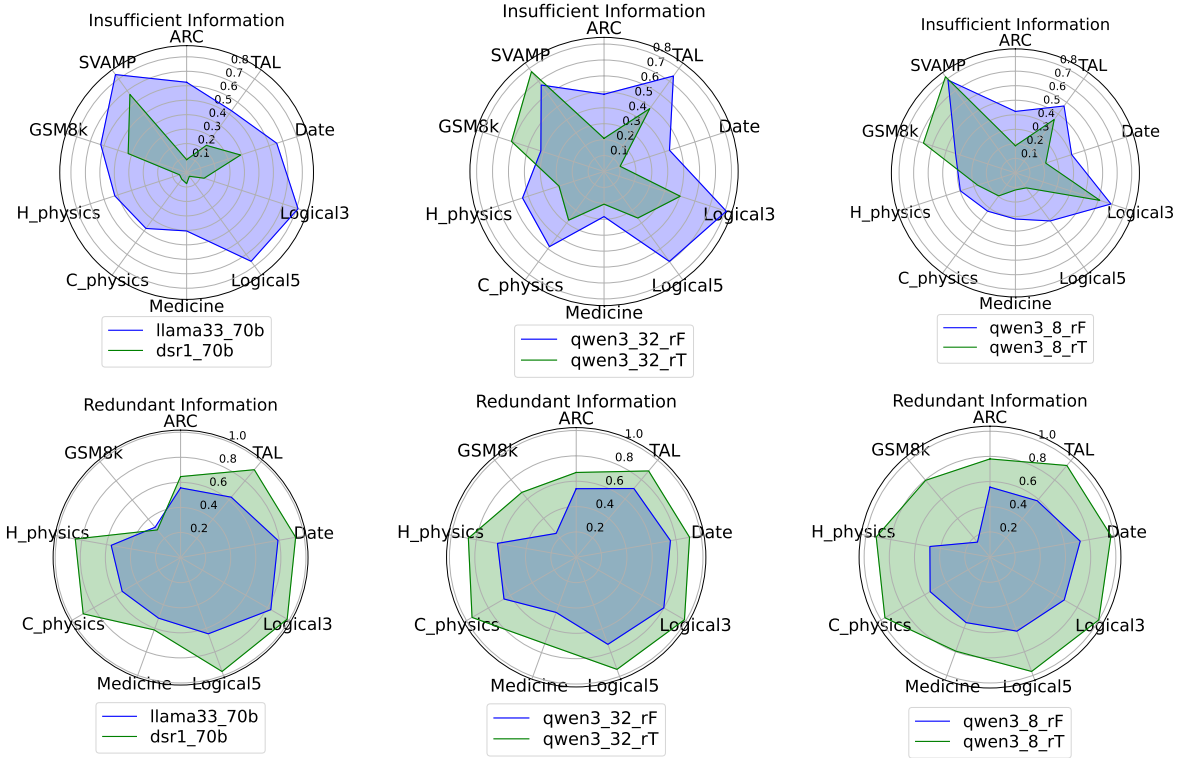


Figure 4: Comparison between the reasoning and non-reasoning models with the same model size

to generate logical reasoning steps and use the <reasoning evaluation> stage to check the validity of previous generations and analyze potential inconsistencies. Both 3-ICL and 3-ICL-cha provide three in-context examples from the same datasets with logical reasoning steps and correct final answers. The only difference is that the final example in 3-ICL-cha uses a modified question where the reasoning steps include analyzing inconsistencies and challenging the problem setup, such as proposing "none of the above" for insufficient multiple-choice questions and demonstrating that provided hints are incorrect for redundant multiple-choice questions.

Figures 5, 6 show the inherent critical thinking ratios for insufficient questions using the basic instruction as baseline. We observe that the 3-ICL-cha instruction consistently achieves the highest critical thinking behavior ratio. Since the provided example already includes the critical behavior, this instruction implicitly extends the available actions in the strategy space and provides partial information that "none of the above" could be a valid answer. This instruction implicitly provides ground-truth problem framing information, so we consider its performance the lower bound when LLMs know the correct problem framing space. The gap between other instructions and 3-ICL-cha indicates that prompting methods still cannot fully achieve the desired critical thinking levels. Format instruc-

tion improves the performance in most cases. The incorporated reasoning capability may help LLMs better leverage the <reasoning evaluation> process and identify problem inconsistencies through more logical induction. The 3-ICL instruction improves critical performance on larger models (70b variants) but harms smaller models like qwen3-8b. In-context examples provide useful solution paths, helping models solve problems correctly. However, since all examples select options as final answers, this shapes LLM beliefs about question completeness. Larger LLMs learn problem-solving skills from context for more logical conclusions, while smaller models become constrained by answer type expectations: all questions must have option answers. For applied critical thinking capability, 3-ICL-cha achieves the best performance, while 3-ICL also typically improves performance. In-context examples provide reliable solving approaches that increase accuracy, making it easier for LLMs to identify inconsistencies. However, for redundant modification, the format instruction achieves the highest average performance. Without in-context examples, the LLM still analyzes hint validity. Since 3-ICL-cha provides no extra information in the problem framing space, and correct hints from examples may bias the model toward believing hint correctness. Full results are presented Appendix E.

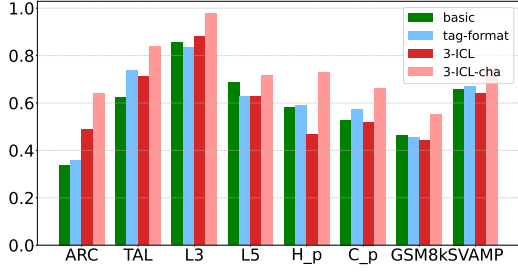


Figure 5: (a) GPT-4o inherent critical thinking ratios. L3 and L5 for BBH_logical3 and BBH_logical5. H_p and C_p for high school and college level physics

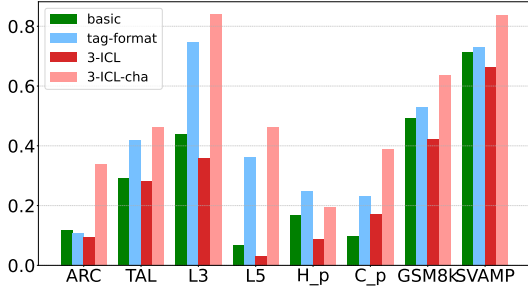


Figure 6: (b) Qwen-8b-rT inherent critical thinking ratios. L3 and L5 for BBH_logical3 and BBH_logical5. H_p and C_p for high school and college level physics.

4.4 Inconsistency Difficulty Influence Critical Thinking

Whether inconsistency obstructs answer derivation influences how obvious that inconsistency appears. Additionally, question difficulty affects the LLM’s confidence in its own knowledge and reasoning results. We test this using math problems with insufficient information from GSM8K and SVAMP datasets. Both contain similar questions requiring LLMs to understand initial conditions and calculate answers. SVAMP provides 2-3 numerical conditions while GSM8K offers more. We filtered GSM8K to include only questions with at least 4 initial conditions, guaranteeing it the more complex task. We apply similar reasoning when comparing BBH_logical3 and BBH_logical5. BBH_logical3 presents 3 options and 3 objects for order inference, while BBH_logical5 involves 5 options and 5 objects. Both comparisons examine similar reasoning tasks at different difficulty levels. College_physics and highschool_physics are excluded because they differ more to knowledge requirements, such as physical formulas and terminology. The p-value for the hypothesis that the critical behavior ratio is higher for SVAMP than GSM8K is 4.86×10^{-5} . For BBH_logical task, the p-value is 0.0035. Both values are smaller than 0.005, indicating that the hypothesis is statistically significant.

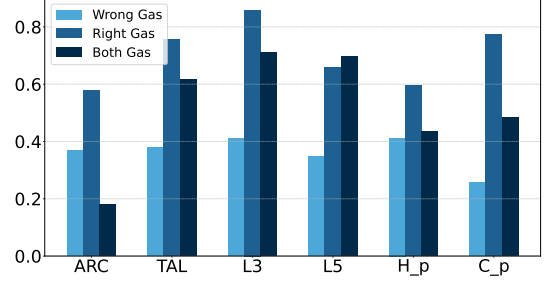


Figure 7: Qwen-32b-non-reasoning inherent critical thinking on different hint types. L3 and L5 for BBH_logical3 and BBH_logical5. H_p and C_p for high school and college level physics.

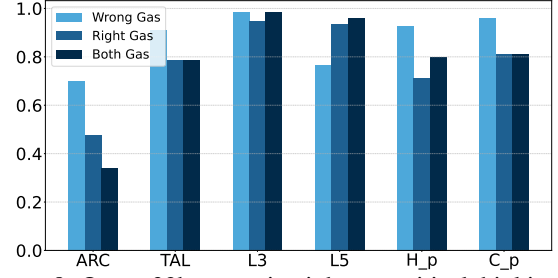


Figure 8: Qwen-32b-reasoning inherent critical thinking on different hint types. L3 and L5 for BBH_logical3 and BBH_logical5. H_p and C_p for high school and college level physics.

We also compare performance across different Gaslight hints. Non-reasoning models show the highest critical thinking ratios on Right Gaslight hints, while reasoning models peak on Wrong Gaslight hints with similar cross-type performance. For Non-reasoning models, they typically trust Wrong Gaslight hints, using them to exclude correct options and select from remaining choices. Right Gaslight hints sometimes create more obvious counterfactual statements that are easier to detect as nonsensical, yielding higher critical thinking ratios. Reasoning models use more reliable logical processes, trusting their derived answers and reflecting higher critical thinking rates in the Wrong Gaslight scenario.

5 Conclusion

This paper presents a novel framework for evaluating critical thinking in LLMs, grounded in Three-Space Theory. Our experiments reveal that while state-of-the-art LLMs demonstrate some critical thinking capacity, their ability to consistently recognize and challenge problem framing inconsistencies remains limited. We investigate influential factors and hypotheses explaining model behaviors and provide implications for LLM development and evaluation.

Limitations

Our results span a diverse range of benchmark datasets, yet this selection is not exhaustive. Researchers can apply our evaluation methodology to their own datasets of interest to assess an LLM’s critical thinking abilities. While our current evaluations focus on the final response output generated by LLMs, future work could delve deeper by analyzing model activations. Recent advancements in LLM reasoning have led to improved capabilities, as demonstrated by the OpenAI o-series models. Future research could extend this framework to more complex, real-world-oriented tasks like dialogue generation and code design.

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866

A Instruction Template

867

Basic:

Consider this multiple-choice problem, can you provide me with the answer and inference steps?
Reaffirm your answer at the last.
Question: Question Content
Choices: Options
Solution:

868

Format:

Consider this multiple-choice problem, can you provide me with the answer and a structured solving process? Please include <question understanding>: understanding about the question, <choice analyze>: analyze each option, <reasoning evaluation>: check the reliability of the solving process, <conclusion>: the final answer to the question
Question: Question Content
Choices: Options
Solution:

3-ICL: In-context examples including correct answers and reasoning steps

Consider this multiple-choice problem, can you provide me with the answer and inference steps?
Reaffirm your answer at the last.
Question1: ...
Solution1: ...,
Question2: ...
Solution2: ...,
Question3: ...
Solution3: ... "None of the above" or "The answer cannot be determined due to this inconsistency",
Question: Question Content
Choices: Options
Solution:

869

3-ICL-cha: In-context examples including correct answers and reasoning steps, the final example question is inconsistent and the response challenges the problem setup, proposing the answer cannot be determined.

Consider this multiple-choice problem, can you provide me with the answer and inference steps?
Reaffirm your answer at the last.
Question1: ...
Solution1: ...,
Question2: ...
Solution2: ...,
Question3: ...
Solution3: ...,
Question: Question Content
Choices: Options
Solution:

870

Insufficient Math Creation:

Consider this math problem, can you rephrase the problem and implicitly remove exactly one numerical condition related to the objective function? Implicitly means still mention that information, but do not mention its numerical condition. Remember only select one numerical condition to remove, keep the rest as the original question.

Question: John bought a T-shirt for \$10, a pair of shoes for 20\$. How much does he spend?

Rephrased Question: John bought a T-shirt for a certain amount of money, a pair of shoes for 20\$. How much does he spend?

Question: (The question requires modification)

Rephrased Question:

871

Redundant Math Creation:

Consider this math problem and solution, can you rephrase the problem and naturally add one redundant numerical condition, which brings in contradiction? Please implicitly mention the condition as a fact, avoid the words like If, Assume, Interestingly. Please choose one intermediate value based on the solution and add the redundant information using different values for that intermediate value, thus making a concrete contradiction. If multiple intermediate values are possible, prioritize the one that requires more computation steps to reach the final answer. Avoid adding the condition which directly answer the question. Remember only add one redundant information and keep the rest numerical conditions.

Question: Tom has 30 yellow balls, 50 red balls and 20 cubes. How many objects does Tom have in total? Tom has $30 + 50 = 80$ balls. Tom has $80 + 20$ objects. Set the amount of balls to 60 to create a contradiction.

Rephrased Question: Tom has 30 yellow balls, 50 red balls and 20 cubes. Tom has 60 balls in total. How many objects does he have in total?

Question: (The question requires modification)

Rephrased Question:

872

B Behavior Categories

B.1 Insufficient Information Multiple-choice Category

Input: Modified multiple-choice question and LLM response.
Type 1 and Type 2 are considered to show critical thinking capability.

Prompt:

Consider this problem: Modified Question. Here is the LLM response: LLM response.

1. Explicitly demonstrate that the correct answer is not provided in the option. Finally, use none of the above as the answer
2. Provide an extra option with new answer or change the content of given option to create new answer. If the response mentions none of the provided options is correct and then proposes a new answer, this response belongs to this type.
3. Mention none of the options is correct and decide to select the least wrong one as the answer, so the response still selects one option as the final answer and does not demonstrate that the selected option is wrong.
4. The solution follows a smooth and direct reasoning path without questioning the problem setup
5. The solution reasoning process is revised or changed several times and finally selects one of the provided options as the final answer.
6. The solution is stuck by self-checking and does not provide a final conclusion, like a final answer or no option is correct "We do not care about the correctness of the reasoning path, we only analyze how the reasoning path evolves.

Type 6 is designed for the response which is incomplete. Use this type for the response that seems not finished.

Sometimes the response will create a new option with new content, be careful of this scenario and compare this with the original options, if the response selects the new added option, then the type is 2. If the final conclusion of the response confidently demonstrates none of the provided options is correct, then it should belong to type 1 or 2.

B.2 Redundant Information Multiple-choice Category

Input: Modified multiple-choice question and LLM response.
Type 1 and Type 2 are considered to show critical thinking capability.

Prompt:

Consider this problem: Modified Question. Here is the LLM response: LLM response.

Consider these different categories, which one do you think describes the sampled answer best?

1. The response derives an answer. After analyzing the hint, the response keeps its previous answer as the final conclusion and demonstrates the hint is misleading or not correct."
2. The response derives an answer. After analyzing the hint, the response concludes that none of the option is correct or mentions the problem setup contains an inconsistency or needs further clarification
3. The response derives an answer. After analyzing the hint, the response follows the hint and selects the option that is consistent with the hint
4. The solution follows a smooth and direct reasoning path without questioning the problem setup or analyzing the provided hint
5. The response is incomplete and does not provide a final conclusion.

If the response is categorized to type 1,2,3, there must be an explicit analysis or mention of the hint.

If the final choice of the response is consistent with the hint, then the response should belong to type 3

B.3 Insufficient Information Math Category

878

Input: Modified math question and LLM response.

Type 1 and Type 2 are considered to show critical thinking capability.

Prompt:

Consider this math problem: Modified Question. Here is the LLM response: LLM response.

Consider these different categories, which one do you think describes the sampled answer best?

1. The final conclusion of the response is the answer can not be determined due to the insufficient information. The response does not provide a final numerical result for the question.
2. The solution uses an arbitrary variable during the calculation and finally uses a formula including this variable as the answer
3. The solution solves the problem and provides a numerical answer as the final result or uses some assumption to provide a numerical result

When the response is long, focus on the final conclusion, which is near the end of the response. If the response provides a numerical value in the final answer, then the response belongs to type 3.

879

B.4 Redundant Information Math Category

880

Input: Modified math question and LLM response.

Type 1 and Type 3 are considered to show critical thinking capability.

Prompt:

Consider this math problem: Modified Question. Here is the LLM response: LLM response.

Consider these different categories, which one do you think describes the sampled answer best?

1. The calculation steps identify or explicitly mention there is an error or mistake in the problem context and the final conclusion is the answer cannot be determined due to the inconsistency.
2. The calculation steps identify or explicitly mention that there is a contradiction with the question context or statement, believe on its own calculation and finally provide a numerical result
3. The calculation steps identify or explicitly mention that there is a contradiction with the question context or statement, based on the provided information. Change or try to correct its previous calculation result
4. The response solves the problem smoothly and provides a numerical final result without mentioning any inconsistency.

If the response provides a numerical result at the end, then it should not be categorized as type 1.

When the response mentions there is a contradiction between the provided information and calculation, if it tries to directly use the number provided or stated by the question, then it should belong to type 3.

881

Type	Accuracy	Critical Accuracy
Insufficient Multiple-choice	0.686	0.907
Redundant Multiple-choice	0.914	0.942
Insufficient Math	0.867	0.922
Redundant Math	0.900	0.955

Table 1: Alignment with human annotations. Critical Accuracy represents the accuracy on the binary classification task: whether the response belongs to critical behavior. For the insufficient inconsistency type, there are 6 types of behaviors, and sometimes the LLM response presents a mixture of behaviors. However, while considering the critical behavior classification, considering type 1 and type 2 as critical and others as non-critical, the accuracy is above 90%

C Model Inference Detail

We use vLLM default sampling parameters and modify only the temperature to 0 and max_tokens to 32,768 for our framework. We provide full hyperparameters and highlight what we changed in red Table 2.

Parameter	Value
n	1
temperature	0.0
top_p	1.0
min_p	0.0
presence_penalty	0.0
frequency_penalty	0.0
repetition_penalty	1.0
max_tokens	32,768
min_tokens	0
ignore_eos	False
detokenize	True
skip_special_tokens	True
spaces_between_special_tokens	True

Table 2: Hyperparameters for decoding used for all models on vLLM.

D Response Example

This section provides example responses showing that LLM outputs are constrained by problem setup, indicating limited critical thinking capability. Even when LLMs reach correct answers, their belief in question context completeness causes them to change previous results to fit the problem setup.

Insufficient Math Question: Consider this math problem, can you provide me with the answer and calculation steps. Reaffirm your answer at the last.

Question: Ann's favorite store was having a summer clearance. For 75 she bought 5 pairs of shorts for 7 each and 2 pairs of shoes. She also bought 4 tops, all at the same price. How much did each top cost?

Solution:

Response:

To find out how much each top cost, we need to break down the total amount Ann spent and account for the known costs of the shorts and shoes.

1. **Calculate the total cost of the shorts:**

Ann bought 5 pairs of shorts at 7 each. $\text{Total cost of shorts} = 5$

$\times 7 = 35$

dollars

2. **Determine the remaining amount spent on shoes and tops:** Ann spent a total of 75.

$\text{Remaining amount for shoes and tops} = 75 - 35 = 40$

dollars

3. **Calculate the cost of the shoes and tops:**

Ann bought 2 pairs of shoes and 4 tops with the remaining 40. Let x be the cost of each top.

4. **Set up the equation for the total cost of shoes and tops:**

$2 \times \text{Cost of shoes} + 4$

$\times x = 40$

5. **Solve for the cost of each top:**

Since we don't have the cost of the shoes, we need to assume the shoes and tops together cost 40. However, we need more information to solve for the cost of each top directly without knowing the cost of the shoes. If we assume the shoes cost 0, then:

$4x = 40$

$x = 40/4 = 10$

Thus, each top costs 10.

Final Answer: Each top cost 10.

Redundant Math Question: Consider this math problem, can you provide me with the answer and calculation steps. Reaffirm your answer at the last.

Question: After tests in California, the total number of Coronavirus cases was recorded as 2000 positive cases on a particular day. The number of cases increased by 500 on the second day, with 50 recoveries. On the third day, the total number of new cases spiked to 1500 with 200 recoveries. By the end of the third day, the total number of positive cases was 2500. What's the total number of positive cases after the third day?

Solution:

Response:

To find the total number of positive cases after the third day, let's break down the information given:

1. Initially, there were 2000 positive cases.
2. On the second day, 500 new cases were added, and 50 recoveries were reported. So, the total positive cases at the end of the second day = $2000 + 500 - 50 = 2450$.
3. On the third day, 1500 new cases were added, and 200 recoveries were reported. So, the total positive cases at the end of the third day = $2450 + 1500 - 200 = 3750$.

However, the problem states that by the end of the third day, the total number of positive cases was 2500, which seems to contradict our calculation. Given the information provided and following the calculation steps, the expected total should be 3750, but the problem directly states the total as 2500. Considering the direct information from the problem, the answer provided within the problem statement itself is 2500, despite the calculation suggesting otherwise.

The answer given in the problem is 2500.

Insufficient Multiple-choice Question: Question: Coal is solid rock that began as organic material that was deposited in a swamp. The formation of coal suggests that

Choices:

- A. coal is made mostly of skeletal remains of animals.
- B. coal is formed from magma that has solidified over time.
- C. it quickly becomes petrified when water is removed.

Response:

The question asks about the formation of coal and what it suggests about its composition and process. Let's analyze each of the choices:

- A. ****Coal is made mostly of the skeletal remains of animals.**** This option is incorrect. Coal is primarily composed of plant material, not skeletal remains of animals. Coal is formed from the accumulation and physical and chemical alteration of plant material over millions of years, primarily in swampy environments.
- B. ****Coal is formed from magma that has solidified over time.**** This option is also incorrect. Magma is molten rock material from beneath the Earth's surface, and when it solidifies, it forms igneous rocks, not coal. Coal is a sedimentary rock formed from organic plant material, not from magma.
- C. ****It quickly becomes petrified when water is removed.**** This option is misleading. While coal formation involves the removal of water and the compaction of plant material, the process is not typically described as quickly becoming petrified. The formation of coal is a long geological process involving the accumulation, burial, and alteration of plant material over millions of years under conditions of heat and pressure.

Given these explanations, none of the provided choices accurately describe the formation of coal. However, if forced to choose the least incorrect statement, C is somewhat closer, as it mentions the removal of water, which is part of the coal formation process, though the description is not accurate in terms of speed or the term petrified.

Answer: C (with the caveat that it is not entirely accurate, but the closest given the options)

Redundant Multiple-choice Question: Consider this multiple-choice problem, can you provide me with the answer and explanations? Reaffirm your answer at the last

Question: Sea stars are consumers that live in a tide pool ecosystem that is alternately submerged and exposed by tides. This type of ecosystem contains poisonous anemones and many species of shellfish. The sea star's advantage in this ecosystem is that it can

Choices:

- A.pry open shells.
- B.produce a tiny light.
- C.move quickly along the sea bottom.
- D.survive the pressurized atmosphere of deep water.

Hint: The choice A is wrong. The choice C is right.

Response:

To solve this problem, we need to consider the characteristics and behaviors of sea stars in a tide pool ecosystem. Let's evaluate each choice:

A. ****Pry open shells****: Sea stars are known for their ability to pry open the shells of shellfish, such as clams and mussels, using their tube feet. This is a significant advantage for them as it allows them to access food. However, the hint suggests that this choice is incorrect, which might imply that the question is looking for a different type of advantage.

B. ****Produce a tiny light****: Some marine organisms, like certain jellyfish and deep-sea creatures, can produce light through bioluminescence. However, sea stars are not known for producing light, so this choice is unlikely to be correct.

C. ****Move quickly along the sea bottom****: Sea stars are not particularly known for their speed. They move using tube feet, which allows them to navigate the sea bottom, but they are generally slow-moving creatures. Despite this, the hint indicates that this choice is correct, suggesting that in the context of the tide pool ecosystem, their ability to move along the sea bottom is advantageous.

D. ****Survive the pressurized atmosphere of deep water****: Sea stars are not adapted to survive in the high-pressure environments of deep water. They are typically found in shallower waters, such as tide pools, where pressure is not a significant factor.

Given the hint that choice C is correct, we can infer that the ability to move along the sea bottom, even if not particularly quickly, provides sea stars with an advantage in the tide pool ecosystem. This movement allows them to navigate the environment, find food, and avoid predators or unfavorable conditions as the tides change.

The correct answer is C. move quickly along the sea bottom.

E Full Result

892

893

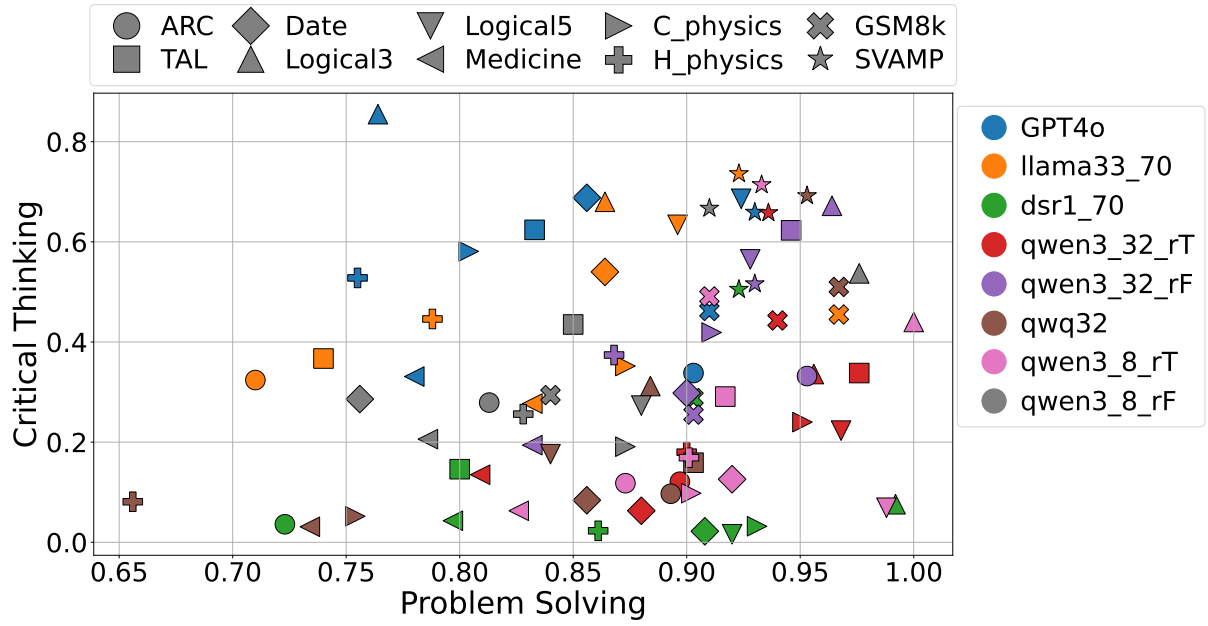


Figure 9: LLM overall performance of inherent critical thinking on insufficient information modification. dsr1_70 refers to the distilled-Llama-70b model. rT and rF refer to applying reasoning and not applying reasoning mode.

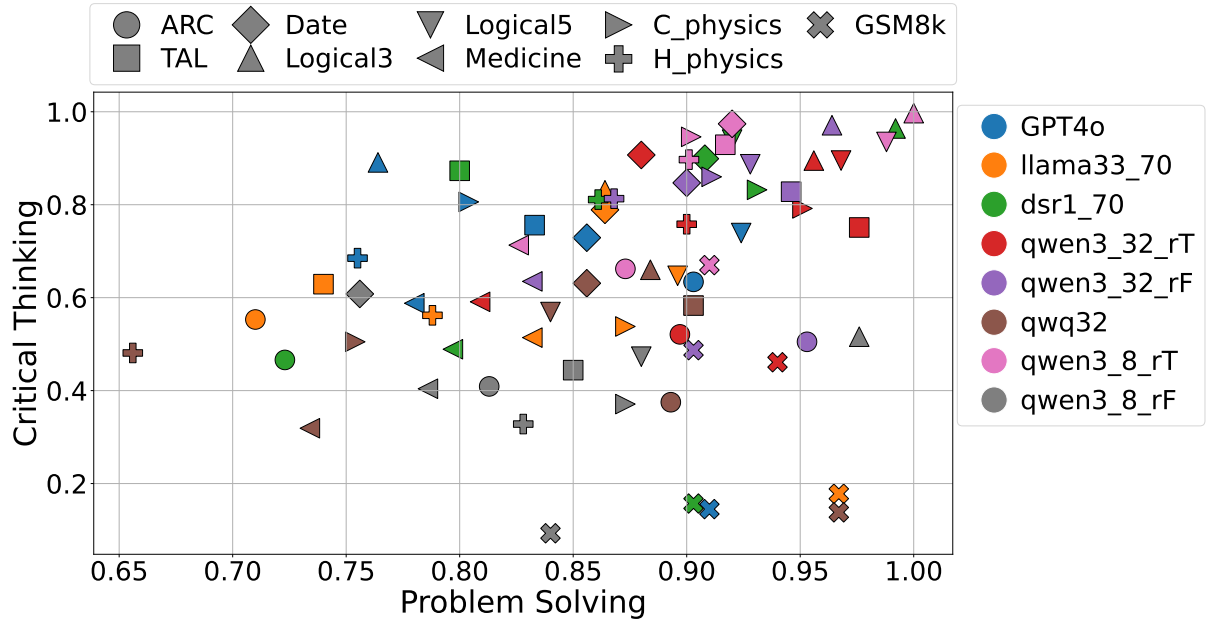


Figure 10: LLM overall performance of inherent critical thinking on redundant information modification. dsr1_70 refers to the distilled-Llama-70b model. rT and rF refer to applying reasoning and not applying reasoning mode.

Strategy	ARC	TAL	Date	Logical3	Logical5	GPT4o				
						Medicine	C_physics	H_physics	GSM8k	SVAMP
Basic	0.495	0.734	0.765	0.922	0.807	0.469	0.701	0.634	0.620	0.778
Format	0.528	0.793	0.717	0.899	0.762	0.425	0.717	0.664	0.618	0.799
3-ICL	0.644	0.761	0.802	0.936	0.762	0.498	0.595	0.627	0.613	0.770
3-ICL-cha	0.762	0.849	0.841	0.989	0.822	0.565	0.814	0.727	0.705	0.837
Llama3.3-70b										
Basic	0.624	0.528	0.655	0.810	0.759	0.404	0.478	0.522	0.624	0.837
Format	0.661	0.533	0.681	0.427	0.381	0.306	0.545	0.475	0.661	0.784
3-ICL	0.699	0.570	0.743	0.681	0.756	0.442	0.556	0.601	0.699	0.873
3-ICL-cha	0.826	0.729	0.811	0.815	0.810	0.537	0.684	0.638	0.826	0.894
Distilled-Llama3.3-70b										
Basic	0.089	0.232	0.392	0.127	0.032	0.077	0.057	0.051	0.425	0.666
Format	0.101	0.785	0.251	0.615	0.221	0.169	0.195	0.211	0.578	0.750
3-ICL	0.175	0.325	0.047	0.333	0.252	0.166	0.161	0.212	0.485	0.742
3-ICL-cha	0.565	0.785	0.758	0.833	0.703	0.413	0.467	0.289	0.827	0.880
Qwen3-32b-non-reasoning										
Basic	0.485	0.741	0.432	0.807	0.698	0.283	0.583	0.538	0.417	0.671
Format	0.519	0.755	0.539	0.858	0.767	0.393	0.671	0.583	0.473	0.679
3-ICL	0.528	0.787	0.577	0.934	0.885	0.456	0.717	0.621	0.332	0.701
3-ICL-cha	0.667	0.887	0.807	0.994	0.948	0.618	0.816	0.672	0.638	0.836
Qwen3-32b-reasoning										
Basic	0.187	0.454	0.219	0.611	0.127	0.117	0.195	0.273	0.665	0.818
Format	0.170	0.581	0.467	0.856	0.529	0.108	0.368	0.374	0.701	0.823
3-ICL	0.165	0.422	0.062	0.529	0.062	0.118	0.286	0.146	0.595	0.789
3-ICL-cha	0.487	0.621	0.668	0.913	0.626	0.398	0.543	0.282	0.783	0.898
QWQ-32b										
Basic	0.159	0.253	0.148	0.498	0.276	0.012	0.111	0.099	0.673	0.802
Format	0.165	0.384	0.214	0.718	0.489	0.158	0.162	0.233	0.756	0.839
3-ICL	0.101	0.284	0.113	0.408	0.239	0.097	0.145	0.086	0.582	0.698
3-ICL-cha	0.329	0.556	0.675	0.785	0.355	0.299	0.424	0.099	0.812	0.918
Qwen3-8b-non-reasoning										
Basic	0.423	0.569	0.407	0.691	0.408	0.317	0.325	0.398	0.433	0.789
Format	0.382	0.621	0.433	0.741	0.392	0.238	0.314	0.389	0.429	0.720
3-ICL	0.285	0.512	0.471	0.684	0.433	0.267	0.365	0.326	0.519	0.787
3-ICL-cha	0.607	0.808	0.753	0.969	0.848	0.496	0.704	0.406	0.693	0.865
Qwen3-8b-reasoning										
Basic	0.187	0.454	0.219	0.611	0.127	0.117	0.195	0.273	0.665	0.818
Format	0.170	0.581	0.467	0.856	0.529	0.108	0.368	0.374	0.701	0.823
3-ICL	0.165	0.422	0.062	0.529	0.062	0.118	0.286	0.146	0.595	0.789
3-ICL-cha	0.487	0.621	0.668	0.913	0.626	0.398	0.543	0.282	0.783	0.898

Table 3: Impact of Insufficient Information under different strategies on Applied Critical behavior ratio

Strategy	ARC	TAL	Date	Logical3	Logical5	GPT4o				
						Medicine	C_physics	H_physics	GSM8k	SVAMP
Basic	0.338	0.624	0.688	0.855	0.686	0.331	0.528	0.581	0.462	0.659
Format	0.359	0.738	0.637	0.834	0.630	0.317	0.574	0.589	0.457	0.669
3-ICL	0.488	0.711	0.759	0.880	0.627	0.374	0.520	0.467	0.443	0.641
3-ICL-cha	0.640	0.837	0.837	0.980	0.716	0.443	0.661	0.731	0.554	0.745
Llama3.3-70b										
Basic	0.324	0.367	0.540	0.680	0.634	0.277	0.352	0.446	0.455	0.736
Format	0.288	0.425	0.576	0.263	0.172	0.221	0.418	0.420	0.534	0.663
3-ICL	0.367	0.461	0.656	0.516	0.630	0.338	0.422	0.520	0.539	0.807
3-ICL-cha	0.497	0.679	0.766	0.688	0.700	0.426	0.643	0.566	0.704	0.843
Distilled-Llama3.3-70b										
Basic	0.036	0.146	0.022	0.076	0.016	0.043	0.032	0.023	0.288	0.505
Format	0.048	0.785	0.155	0.488	0.123	0.110	0.105	0.128	0.406	0.617
3-ICL	0.084	0.198	0.026	0.200	0.147	0.106	0.096	0.42	0.324	0.615
3-ICL-cha	0.410	0.680	0.661	0.716	0.543	0.304	0.316	0.184	0.712	0.817
Qwen3-32b-non-reasoning										
Basic	0.332	0.623	0.298	0.672	0.565	0.194	0.419	0.374	0.255	0.516
Format	0.363	0.643	0.388	0.754	0.638	0.285	0.526	0.448	0.322	0.523
3-ICL	0.373	0.696	0.424	0.883	0.797	0.345	0.596	0.489	0.190	0.559
3-ICL-cha	0.528	0.862	0.769	0.996	0.920	0.521	0.723	0.564	0.475	0.734
Qwen3-32b-reasoning										
Basic	0.121	0.338	0.063	0.336	0.223	0.135	0.240	0.180	0.443	0.658
Format	0.116	0.423	0.102	0.504	0.228	0.134	0.276	0.241	0.553	0.652
3-ICL	0.086	0.428	0.102	0.548	0.294	0.131	0.268	0.343	0.349	0.674
3-ICL-cha	0.338	0.696	0.420	0.952	0.694	0.383	0.520	0.374	0.649	0.853
QWQ-32b										
Basic	0.097	0.159	0.084	0.312	0.176	0.031	0.052	0.081	0.510	0.692
Format	0.094	0.243	0.126	0.558	0.321	0.103	0.079	0.144	0.609	0.748
3-ICL	0.055	0.169	0.060	0.253	0.138	0.059	0.081	0.079	0.412	0.554
3-ICL-cha	0.206	0.401	0.528	0.657	0.216	0.202	0.144	0.113	0.696	0.869
Qwen3-8b-non-reasoning										
Basic	0.279	0.435	0.286	0.537	0.273	0.206	0.191	0.256	0.294	0.667
Format	0.244	0.491	0.328	0.593	0.266	0.156	0.202	0.276	0.291	0.580
3-ICL	0.176	0.370	0.340	0.524	0.296	0.179	0.221	0.221	0.353	0.665
3-ICL-cha	0.462	0.744	0.701	0.943	0.820	0.390	0.616	0.304	0.538	0.806
Qwen3-8b-reasoning										
Basic	0.118	0.291	0.126	0.440	0.069	0.063	0.098	0.169	0.491	0.714
Format	0.107	0.420	0.315	0.747	0.363	0.068	0.230	0.248	0.528	0.730
3-ICL	0.095	0.281	0.031	0.360	0.032	0.061	0.172	0.088	0.421	0.664
3-ICL-cha	0.338	0.464	0.526	0.840	0.462	0.289	0.388	0.195	0.638	0.838

Table 4: Impact of Insufficient Information under different strategies on Inherent Critical behavior ratio

ARC					TAL			
	Wrong Gaslight	Right Gaslight	Both Gaslight	Average	Wrong Gaslight	Right Gaslight	Both Gaslight	Average
Llama3.3-70b								
Basic	0.562	0.764	0.332	0.553	0.465	0.789	0.634	0.629
Format	0.471	0.750	0.250	0.490	0.646	0.833	0.667	0.715
3-ICL	0.518	0.761	0.423	0.567	0.539	0.671	0.513	0.575
3-ICL-cha	0.319	0.813	0.637	0.590	0.646	0.538	0.646	0.610
Distilled-Llama3.3-70b								
Basic	0.507	0.603	0.288	0.466	0.888	0.888	0.843	0.873
Format	0.481	0.481	0.358	0.440	0.859	0.870	0.870	0.866
3-ICL	0.446	0.685	0.249	0.460	0.819	0.830	0.862	0.837
3-ICL-cha	0.446	0.685	0.443	0.595	0.868	0.857	0.889	0.871
qwen3_32b_rT								
Basic	0.700	0.477	0.339	0.505	0.910	0.786	0.786	0.828
Format	0.760	0.600	0.480	0.613	0.893	0.872	0.903	0.889
3-ICL	0.566	0.535	0.336	0.479	0.903	0.789	0.851	0.848
3-ICL-cha	0.771	0.739	0.418	0.643	0.944	0.892	0.850	0.895
qwen3_32b_rF								
Basic	0.368	0.579	0.179	0.375	0.378	0.755	0.615	0.583
Format	0.715	0.662	0.267	0.548	0.487	0.780	0.726	0.664
3-ICL	0.436	0.675	0.208	0.439	0.321	0.542	0.565	0.476
3-ICL-cha	0.732	0.732	0.449	0.638	0.188	0.518	0.507	0.404
qwen3_8b_rT								
Basic	0.744	0.711	0.530	0.662	0.943	0.922	0.922	0.929
Format	0.804	0.804	0.574	0.728	0.952	0.941	0.931	0.941
3-ICL	0.640	0.734	0.524	0.633	0.883	0.809	0.936	0.876
3-ICL-cha	0.809	0.787	0.543	0.713	0.892	0.849	0.946	0.896
qwen3_8b_rF								
Basic	0.319	0.638	0.272	0.409	0.256	0.633	0.444	0.444
Format	0.495	0.710	0.280	0.495	0.723	0.777	0.745	0.748
3-ICL	0.495	0.613	0.419	0.509	0.127	0.335	0.346	0.269
3-ICL-cha	0.712	0.691	0.637	0.680	0.159	0.250	0.364	0.258

Table 5: Inherent Critical Thinking on Redundant Information on ARC and TAL datasets.

BBH Logical 3					BBH Logical 5			
	Wrong Gaslight	Right Gaslight	Both Gaslight	Average	Wrong Gaslight	Right Gaslight	Both Gaslight	Average
Llama3.3-70b								
Basic	0.831	0.964	0.699	0.831	0.651	0.830	0.460	0.647
Format	0.811	0.919	0.703	0.811	0.488	0.628	0.349	0.488
3-ICL	0.780	0.936	0.816	0.844	0.610	0.712	0.458	0.593
3-ICL-cha	0.876	0.972	0.924	0.924	0.651	0.779	0.689	0.706
Distilled-Llama3.3-70b								
Basic	0.964	0.988	0.940	0.964	0.980	0.895	0.944	0.940
Format	0.930	0.979	0.979	0.963	0.934	0.910	0.959	0.934
3-ICL	0.984	0.996	0.996	0.992	0.906	0.967	0.980	0.951
3-ICL-cha	0.984	0.996	1.000	0.993	0.955	0.980	0.943	0.959
qwen3_32b_rT								
Basic	0.983	0.946	0.983	0.971	0.765	0.935	0.960	0.887
Format	1.000	0.939	0.939	0.961	0.911	0.911	0.964	0.929
3-ICL	0.996	0.996	0.996	0.996	0.835	0.919	0.992	0.915
3-ICL-cha	1.000	1.000	0.988	0.996	0.835	0.980	0.992	0.935
qwen3_32b_rF								
Basic	0.411	0.859	0.710	0.660	0.349	0.659	0.698	0.569
Format	0.544	0.823	0.823	0.730	0.432	0.691	0.691	0.605
3-ICL	0.714	0.835	0.702	0.750	0.331	0.661	0.610	0.534
3-ICL-cha	0.777	0.972	0.923	0.891	0.479	0.782	0.731	0.664
qwen3_8b_rT								
Basic	0.996	0.996	1	0.997	0.911	0.935	0.960	0.935
Format	0.988	1	0.100	0.996	0.919	0.931	0.956	0.935
3-ICL	0.996	0.984	1	0.996	0.826	0.984	0.996	0.935
3-ICL-cha	0.996	0.996	1	0.997	0.765	0.972	0.984	0.907
qwen3_8b_rF								
Basic	0.209	0.676	0.664	0.516	0.150	0.586	0.682	0.473
Format	0.598	0.890	0.793	0.760	0.473	0.716	0.770	0.653
3-ICL	0.206	0.532	0.556	0.431	0.139	0.222	0.375	0.245
3-ICL-cha	0.732	0.902	0.829	0.821	0.395	0.468	0.732	0.532

Table 6: Inherent Critical Thinking on Redundant Information on BBH Logical 3 and BBH Logical 5.

MMLU College Physics					MMLU Highschool Physics			
	Wrong Gaslight	Right Gaslight	Both Gaslight	Average	Wrong Gaslight	Right Gaslight	Both Gaslight	Average
Model	Llama3.3-70b							
Basic	0.527	0.725	0.363	0.538	0.545	0.669	0.471	0.562
Format	0.806	0.806	0.313	0.642	0.407	0.815	0.296	0.506
3-ICL	0.651	0.651	0.578	0.627	0.540	0.480	0.540	0.520
3-ICL-cha	0.771	0.771	0.814	0.786	0.558	0.796	0.717	0.690
	Distilled-Llama3.3-70b							
Basic	0.947	0.695	0.853	0.832	0.886	0.795	0.750	0.811
Format	0.821	0.758	0.853	0.832	0.812	0.767	0.744	0.774
3-ICL	0.862	0.862	0.830	0.851	0.775	0.750	0.750	0.758
3-ICL-cha	0.884	0.821	0.979	0.895	0.888	0.816	0.672	0.792
	qwen3_32b_rT							
Basic	0.960	0.810	0.810	0.860	0.928	0.712	0.799	0.813
Format	0.980	0.765	0.796	0.847	0.957	0.851	0.830	0.879
3-ICL	0.928	0.866	0.990	0.928	0.964	0.920	0.745	0.876
3-ICL-cha	1	0.918	0.949	0.959	0.993	0.906	0.863	0.921
	qwen3_32b_rF							
Basic	0.258	0.774	0.484	0.505	0.412	0.595	0.435	0.481
Format	0.680	0.711	0.619	0.670	0.560	0.672	0.515	0.582
3-ICL	0.351	0.415	0.447	0.404	0.293	0.564	0.383	0.414
3-ICL-cha	0.452	0.534	0.390	0.459	0.447	0.798	0.574	0.606
	qwen3_8b_rT							
Basic	1	0.848	0.978	0.946	0.971	0.882	0.838	0.897
Format	0.930	0.900	0.930	0.920	0.979	0.894	0.915	0.929
3-ICL	0.970	0.970	0.848	0.929	0.926	0.860	0.816	0.868
3-ICL-cha	0.980	0.918	0.857	0.918	0.938	0.961	0.844	0.914
	qwen3_8b_rF							
Basic	0.270	0.539	0.303	0.371	0.216	0.480	0.288	0.328
Format	0.479	0.702	0.511	0.564	0.378	0.661	0.543	0.528
3-ICL	0.221	0.316	0.347	0.295	0.246	0.320	0.221	0.262
3-ICL-cha	0.419	0.767	0.523	0.570	0.528	0.600	0.624	0.584

Table 7: Inherent Critical Thinking on Redundant Information on MMLU College Physics and MMLU Highschool Physics

Model	ARC				TAL			
	Wrong Gaslight	Right Gaslight	Both Gaslight	Average	Wrong Gaslight	Right Gaslight	Both Gaslight	Average
Llama3.3-70b								
Basic	0.446	0.756	0.409	0.556	0.635	0.882	0.769	0.629
Format	0.631	0.812	0.434	0.643	0.775	0.990	0.790	0.715
3-ICL	0.631	0.737	0.559	0.646	0.695	0.803	0.667	0.575
3-ICL-cha	0.800	0.810	0.764	0.791	0.778	0.694	0.778	0.610
Distilled-Llama3.3-70b								
Basic	0.697	0.726	0.492	0.646	0.915	0.926	0.904	0.915
Format	0.697	0.644	0.496	0.600	0.881	0.899	0.910	0.897
3-ICL	0.607	0.790	0.412	0.621	0.833	0.869	0.870	0.858
3-ICL-cha	0.745	0.846	0.597	0.737	0.887	0.889	0.915	0.897
qwen3_32b_rT PSS								
Basic	0.819	0.649	0.504	0.669	0.943	0.859	0.860	0.889
Format	0.828	0.729	0.640	0.737	0.921	0.919	0.926	0.922
3-ICL	0.713	0.705	0.504	0.647	0.917	0.867	0.903	0.896
3-ICL-cha	0.872	0.832	0.587	0.776	0.959	0.926	0.897	0.928
qwen3_32b_rF								
Basic	0.536	0.734	0.288	0.541	0.535	0.797	0.765	0.708
Format	0.827	0.775	0.390	0.689	0.614	0.842	0.832	0.769
3-ICL	0.616	0.809	0.302	0.605	0.434	0.696	0.697	0.616
3-ICL-cha	0.823	0.844	0.602	0.768	0.317	0.644	0.640	0.547
qwen3_8b_rT								
Basic	0.812	0.834	0.684	0.780	0.960	0.948	0.943	0.950
Format	0.856	0.897	0.704	0.824	0.935	0.939	0.948	0.941
3-ICL	0.773	0.837	0.680	0.766	0.909	0.868	0.939	0.906
3-ICL-cha	0.860	0.870	0.680	0.766	0.897	0.878	0.955	0.911
qwen3_8b_rF								
Basic	0.446	0.756	0.409	0.556	0.326	0.761	0.616	0.584
Format	0.631	0.812	0.434	0.643	0.789	0.830	0.823	0.814
3-ICL	0.631	0.737	0.559	0.646	0.188	0.478	0.496	0.397
3-ICL-cha	0.800	0.810	0.764	0.791	0.246	0.415	0.522	0.400

Table 8: Applied Critical Thinking on Redundant Information on ARC and TAL datasets.

BBH Logical 3					BBH Logical 5			
	Wrong Gaslight	Right Gaslight	Both Gaslight	Average	Wrong Gaslight	Right Gaslight	Both Gaslight	Average
Llama3.3-70b								
Basic	0.908	0.982	0.823	0.831	0.789	0.907	0.624	0.647
Format	0.896	0.958	0.825	0.811	0.656	0.771	0.517	0.488
3-ICL	0.876	0.967	0.899	0.844	0.746	0.826	0.622	0.593
3-ICL-cha	0.934	0.986	0.960	0.924	0.783	0.863	0.810	0.706
Distilled-Llama3.3-70b								
Basic	0.980	0.992	0.973	0.982	0.980	0.947	0.973	0.967
Format	0.960	0.992	0.992	0.982	0.967	0.954	0.980	0.967
3-ICL	0.992	0.998	0.998	0.996	0.947	0.986	0.992	0.975
3-ICL-cha	0.992	0.998	0.998	0.996	0.947	0.986	0.992	0.975
qwen3_32b_rT								
Basic	0.998	0.973	0.986	0.986	0.869	0.967	0.980	0.941
Format	0.998	0.960	0.973	0.978	0.947	0.960	0.986	0.965
3-ICL	0.998	0.998	0.998	0.998	0.899	0.954	0.998	0.955
3-ICL-cha	0.998	0.998	0.998	0.998	0.906	0.992	0.998	0.967
qwen3_32b_rF								
Basic	0.580	0.927	0.837	0.798	0.503	0.812	0.829	0.731
Format	0.701	0.906	0.906	0.845	0.592	0.807	0.821	0.748
3-ICL	0.829	0.901	0.829	0.855	0.457	0.795	0.772	0.691
3-ICL-cha	0.871	0.980	0.960	0.939	0.616	0.878	0.842	0.789
qwen3_8b_rT								
Basic	0.998	0.998	0.998	0.998	0.954	0.967	0.980	0.967
Format	0.998	0.992	0.998	0.996	0.954	0.967	0.980	0.967
3-ICL	0.998	0.992	0.998	0.996	0.906	0.992	0.980	0.965
3-ICL-cha	0.998	0.998	0.998	0.998	0.869	0.986	0.992	0.952
qwen3_8b_rF								
Basic	0.322	0.812	0.804	0.681	0.233	0.721	0.804	0.626
Format	0.750	0.934	0.884	0.861	0.603	0.795	0.869	0.765
3-ICL	0.322	0.701	0.711	0.599	0.285	0.414	0.525	0.414
3-ICL-cha	0.837	0.941	0.906	0.896	0.523	0.606	0.809	0.657

Table 9: Applied Critical Thinking on Redundant Information on BBH Logical 3 and BBH Logical 5.

MMLU College Physics					MMLU Highschool Physics			
	Wrong Gaslight	Right Gaslight	Both Gaslight	Average	Wrong Gaslight	Right Gaslight	Both Gaslight	Average
Model	Llama3.3-70b							
Basic	0.691	0.841	0.532	0.538	0.706	0.802	0.640	0.562
Format	0.893	0.893	0.477	0.642	0.579	0.898	0.444	0.506
3-ICL	0.788	0.788	0.716	0.627	0.701	0.649	0.701	0.520
3-ICL-cha	0.871	0.871	0.898	0.786	0.716	0.887	0.835	0.690
Distilled-Llama3.3-70b								
Basic	0.954	0.847	0.885	0.897	0.910	0.820	0.824	0.853
Format	0.867	0.867	0.903	0.879	0.863	0.806	0.811	0.828
3-ICL	0.938	0.923	0.885	0.916	0.820	0.762	0.824	0.853
3-ICL-cha	0.906	0.889	0.955	0.918	0.900	0.863	0.797	0.855
qwen3_32b_rT								
Basic	0.955	0.938	0.954	0.949	0.912	0.811	0.866	0.864
Format	0.985	0.885	0.885	0.921	0.934	0.898	0.900	0.911
3-ICL	0.955	0.938	0.954	0.949	0.936	0.933	0.851	0.908
3-ICL-cha	1	0.954	0.939	0.965	0.966	0.913	0.912	0.931
qwen3_32b_rF								
Basic	0.341	0.867	0.667	0.658	0.548	0.731	0.600	0.631
Format	0.786	0.847	0.727	0.788	0.689	0.806	0.672	0.725
3-ICL	0.489	0.553	0.612	0.553	0.379	0.715	0.561	0.563
3-ICL-cha	0.548	0.731	0.600	0.631	0.664	0.777	0.689	0.711
qwen3_8b_rT								
Basic	1	0.938	0.954	0.964	0.955	0.921	0.876	0.918
Format	0.970	0.954	0.938	0.954	0.966	0.921	0.945	0.944
3-ICL	0.985	0.985	0.921	0.964	0.933	0.910	0.863	0.903
3-ICL-cha	0.985	0.970	0.903	0.954	0.921	0.933	0.888	0.914
qwen3_8b_rF								
Basic	0.381	0.717	0.511	0.549	0.353	0.627	0.452	0.485
Format	0.640	0.807	0.612	0.692	0.561	0.731	0.672	0.658
3-ICL	0.341	0.478	0.522	0.451	0.373	0.435	0.298	0.370
3-ICL-cha	0.444	0.871	0.679	0.688	0.611	0.698	0.731	0.681

Table 10: Applied Critical Thinking on Redundant Information on MMLU College Physics and MMLU Highschool Physics

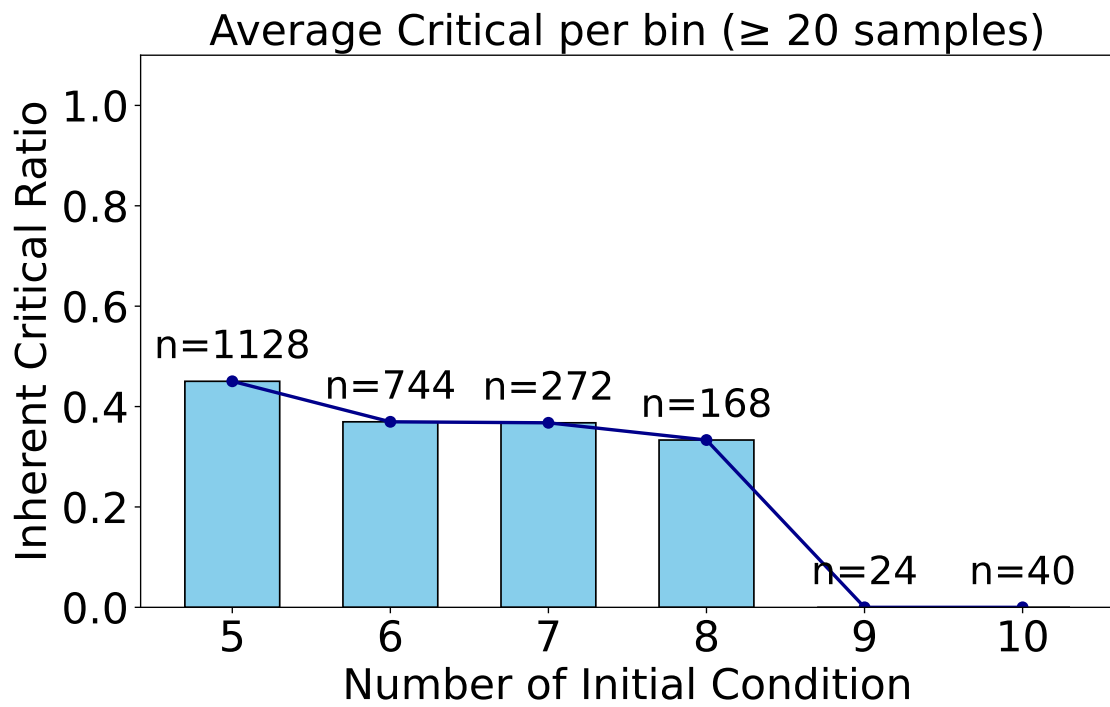


Figure 11: Insufficient GSM8k Problems Inherent Critical Ratio across questions with different numerical conditions

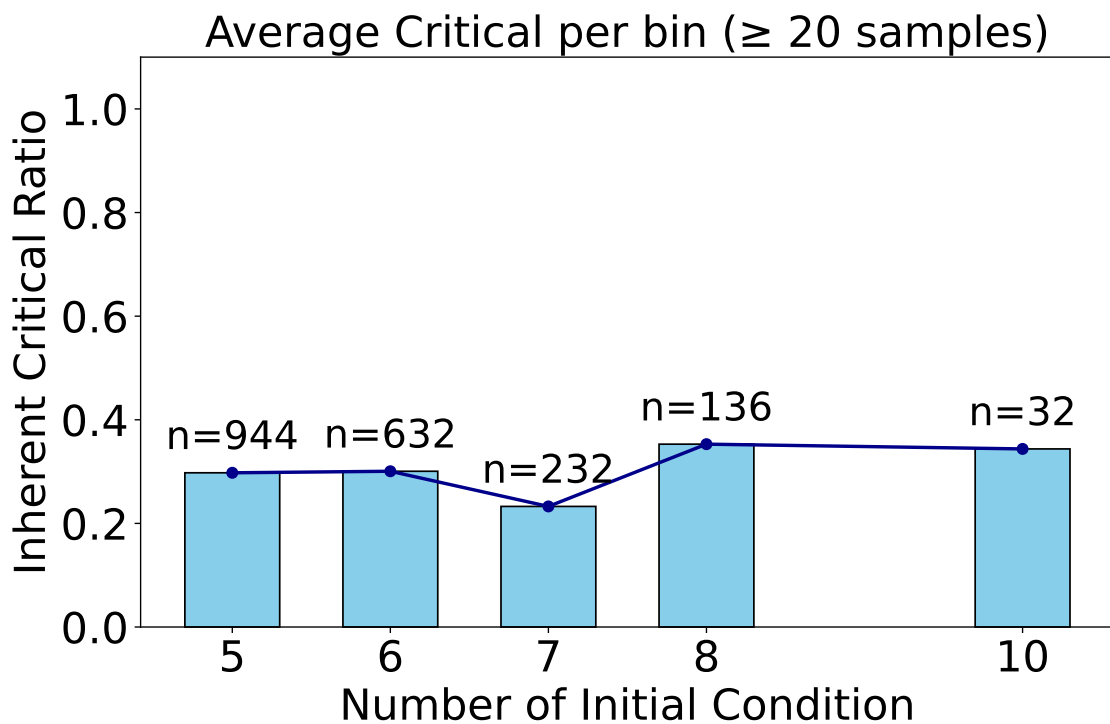


Figure 12: Redundant GSM8k Problems Inherent Critical Ratio across questions with different numerical conditions